Exercises Week 3

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Exercise 1

Let $X(t)$ be a Gaussian zero-mean process with t $\in \mathbf{R}^d$, d = 1,2, and with Matérn covariance function.

- Let's draw a realization of one-dimensional Matérn with $\nu = 1.75$

Let's use the following discretisation

t = ih, h = 0.01, i = 0,...,100 in dimension one

$$t_i \in [0, 1] \tag{1}$$

The distance D_{ij}

$$D_{ij} = ||t_i - t_j||_2 \tag{2}$$

The Matérn covariance

$$\begin{aligned} C_v(d_{ij}) &= \sigma^2 \frac{2^{1-v}}{\Gamma(v)} \left(\frac{\sqrt{2v}d}{l} \right)^v K_v \left(\frac{\sqrt{2v}d}{l} \right), i \neq j \\ C_v(d_{ii}) &= \sigma_f^2, \forall i \in \{1, ..., n\} \end{aligned} \tag{3}$$

(Rasmussen, 2005)

Let's joint-estimate σ_n and the states. One run is done where the prior of $\sigma_n \sim U(0.05, 1.35)$ and in the second run the prior is $\sigma_n \sim$ Log-Normal(log 0.2, 0.5^2). The true values are $\sigma_f = 1, l = 0.45, \nu = 0.45$.

The observed data are $y = x + e, e \sim N(0, \sigma_n I)$. The ground truth is a single from the Matérn GP prior.

Now the Matérn prior smoothness ν, σ_n and the states. In the first run both are uniformly distributed and in the second run both are log-normally distributed. σ_n follows the distribution as defined before whereas the uniform distribution of nu is $\nu \sim U(0.1, 5)$ and its log-normal distribution is $\nu \sim log - normal(0.5, 0.25^2)$

Let C(ν) = $L(\nu)L(\nu)^T$ be the Cholesky factorization and let's note $x = L(\nu)u$ with u $\sim N(0, I)$. The prior of u is independent of ν. The posterior is:

$$\pi(u, \sigma_n, \nu|y) \propto \exp(-\Phi(u; \sigma_n, \nu))\sigma_n^{-N} N(u; 0, I)p(\sigma_n)p(\nu) \tag{4}$$

and the negative log-likelihood is:

$$\Phi(u; \sigma_n, v) = \frac{1}{2\sigma_n^2}||y - L(\nu)||^2 \tag{5}$$

To sample we use pCn to propose a new state: $u' = \sqrt{1-\beta^2}u + \beta\xi, \xi \sim N(0, I)$. The acceptance happens with probability: $\min\{1, \exp(\Phi(u) - \Phi(u')\}$. In this case $\beta = 0.15$. σ_n has a such proposal: $\log \sigma_n^{'} = \log \sigma_n + 0.05z, z \sim N(0.1)$ and the proposal is accepted with:

$$\alpha = \min\{1, \exp\exp(\Phi - \Phi^{'} \left(\frac{\sigma_n}{\sigma_n^{'}}\right) \frac{p(\sigma_n^{'})}{p(\sigma_n)} \frac{\sigma_n^{'}}{\sigma_n}\} \tag{6}$$

, the last factor is just the Jacobian of the log transform. In the task 1.2, where ν is also estimated we propose $\nu^{'} = \nu + U(-0.1, 0.1)$. The acceptance is:

$$\alpha = \min\{1, \exp(\Phi(u; v) - \Phi^{'}(u; \nu^{'}) \left(\frac{\sigma_n}{\sigma_n^{'}}\right) \frac{p(\nu)}{p(\nu^{'})}\} \tag{7}$$

(Dashti and Stuart, 2015).
I run 30 000 iterations and first 5000 were the burn-in.
The results for uniform and log-normal for σ_n

	Uniformly dist.	Log-norm
Acceptance Rate	78.13%	78.17%

Initial values for the second run are $\nu^{(0)} = 0.5$ and $\sigma_n^{(0)} = 0.1$. The results for uniform and log-normal for σ_n

	Uniformly dist.	Log-norm
σ_n acceptance rate	78.55%	78.17%
ν acceptance rate	45.53%	49.95%

Exercise 2

In the second task we still have the Matérn prior, non-centred parametrisation, pCN update of the state and random walk updates of σ_n. However this time:

The state is not directly observed but rather

$$y(t) = \int K(t - t^{'})x(t^{'})dt^{'} + e(t) \rightarrow y = Kx + e, e \sim N(0, \sigma_n^2 I) \tag{8}$$

. K representes a forward operator. In fact, it is a boxcar kernel.

$$K(\tau) = 1\{-0.2 \leq \tau \leq 0.2\} \tag{9}$$

(Kaipio and Somersalo, 2007).

The negative log-likehood becomes:

$$\Phi(u; \sigma_n, v) = \frac{1}{2\sigma_n^2}||y - KL(\nu)||^2 \tag{10}$$

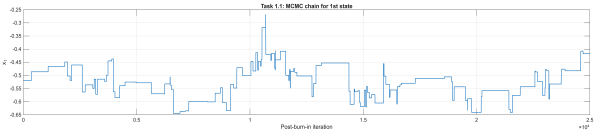
The truth is this time an analytical function:

$$x_{true}(t) = 0.8\sin(2\pi t) + 0.4\cos(3\pi t) + 0.5\exp(-50(t - 0.65)^2) \tag{11}$$

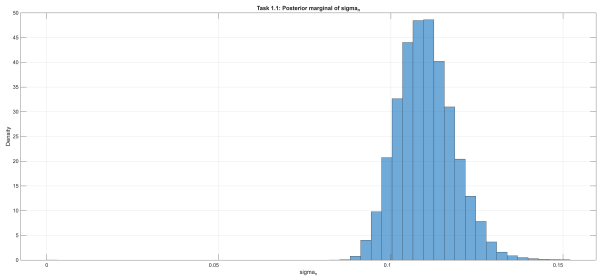
	Uniformly dist.	Log-norm
σ_n Acceptance Rate	78.23%	78.25%

Initial values for the second run are $\nu^{(0)} = 0.5$ and $\sigma_n^{(0)} = 0.1$. The results for uniform and log-normal for σ_n

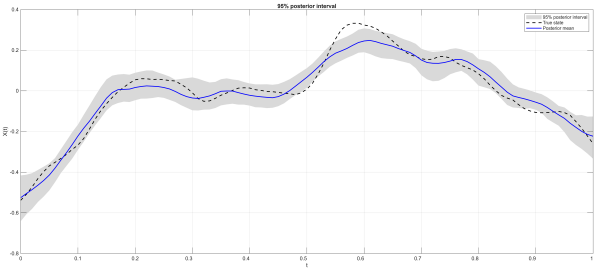
	Uniformly dist.	Log-norm
σ_n acceptance rate	78.54%	78.54%
ν acceptance rate	88.97%	87.35%



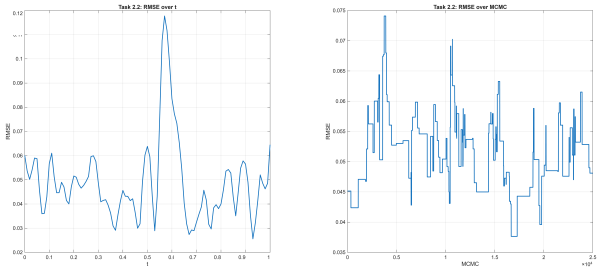
(a) Task 1.1 results



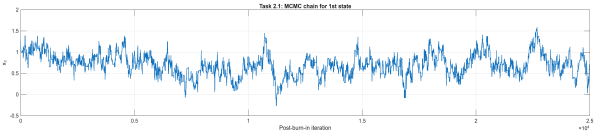
(b) Task 1.1 MCMC



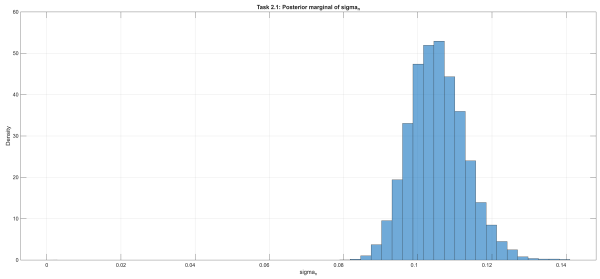
(c) Task 1.1 posterior



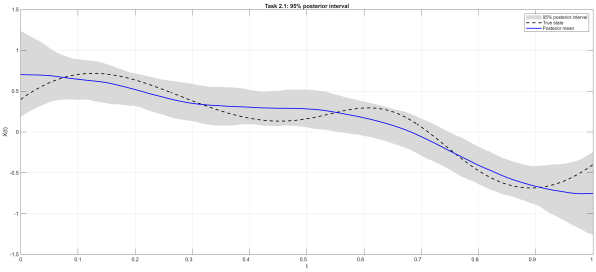
(d) Task 1.1 RMSE



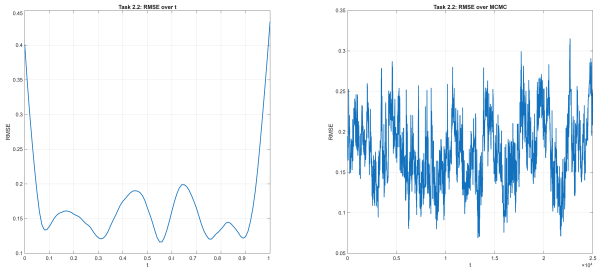
(e) Task 2.1 results



(f) Task 2.1 MCMC



(g) Task 2.1 posterior



(h) Task 2.1 RMSE

References

Dashti, M. and Stuart, A. M. (2015). The bayesian approach to inverse problems.

Kaipio, J. and Somersalo, E. (2007). Statistical inverse problems: Discretization, model reduction and inverse crimes. *Journal of Computational and Applied Mathematics*, 198(2):493–504. Special Issue: Applied Computational Inverse Problems.

Rasmussen, Carl Edward, W. C. K. (2005). *Gaussian Processes for Machine Learning*. MIT Press.